Department of the Interior and Local Government
Regional Office XIII — Caraga

Local Government Monitoring and Evaluation Division

PROPOSAL

An AI Assistant for LGMED-IMMS

Adding a built-in assistant, powered by a large language model (LLM), that answers the public's questions from the information the Division has already published

System	LGMED-IMMS (Information Management and Monitoring System)
Document	Proposal and briefing manual for the LGMED team
Version	1.0 — for discussion
Date	October 2026
Prepared by	

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1. Summary

The proposal in one paragraph

We add an **AI Assistant** to the public website of LGMED-IMMS. A visitor (a citizen, an LGU staff member, a partner agency) types a question in their own words, such as *"What programs does LGMED run in Agusan del Sur?"* or *"Paano mag-request ng dokumento?"* The assistant looks through the information the Division **has already published** on the site (announcements, programs, frontline services, accomplishments, statistics, reports, documents and the calendar) and replies straight away in plain language, with links to the pages it used.

Why do this

- **Visitors find answers faster.** They don't have to know which page to open; they just ask.
- **Fewer routine enquiries.** Questions like "what are your office hours", "what are the requirements for…" and "when is the next activity" are answered without a staff member having to reply.
- **Answers come from our own records.** The assistant only uses what the Division has approved for publication, and every answer links to its source, so anyone can check it.
- **Always up to date.** When a record is published or changed in IMMS, the assistant uses the new version. Nobody has to re-train or re-teach it.
- **It builds on what we have.** The public website and its data already exist in IMMS. The assistant is a new feature on top of them, not a new system.

What we recommend

1. Start with a **public assistant that answers only from published information**.
2. Run it on a **hosted AI service** first (Option A in Section 6). This needs no new hardware, gives the best answer quality, and is cheap at the volume we expect. Because the assistant only reads information that is *already public*, sending it to an outside service exposes nothing new.
3. Build it so the AI engine can be **swapped**. If the team later wants an assistant for *internal* records, which must stay in the office, we buy a local AI workstation (Section 7) and switch to it without rebuilding.
4. Run a **four-week staff-only pilot** before the assistant goes public.

2. What an LLM is, in plain terms

A **large language model (LLM)** is the technology behind tools such as ChatGPT, Claude and Gemini. It has read a very large amount of text and learned how language works, so it can understand a question written in ordinary words and write a clear answer.

On its own, an LLM knows a lot about the world in general but **nothing about our Division**: not our programs, our LGUs, our announcements or our schedule. If you ask it about them without giving it our information, it may guess, and a guess can sound confident and still be wrong. That is called a **"hallucination"**.

How we make it answer from our data

We do **not** re-train the model on our data. That is expensive, slow, and would leave it out of date the moment a record changes. We use a method called **retrieval-augmented generation (RAG)**:

1. When a question comes in, IMMS first **searches our published records** for the passages most likely to answer it.
2. IMMS hands those passages to the LLM together with the question and a strict instruction: *"Answer only from these passages. If they do not contain the answer, say so and point the person to the contact page."*
3. The LLM writes the answer and IMMS attaches links to the pages it used.

> **An easy way to explain it:** the LLM is like a capable new staff member who writes and speaks well but has never worked at LGMED. RAG is handing them the right folder before each question and telling them to answer only from what is in it.

3. What the assistant will do

For the public (Phase 1 — this proposal)

A chat button appears on every page of the public website. A visitor clicks it, types a question, and gets an answer in a few seconds. The visitor does not need an account.

Example conversation

Visitor

What programs does LGMED have for barangays?

LGMED AI Assistant

LGMED currently publishes three programs that involve barangays:
• **[Program A]** — assessment of barangay governance performance…
• **[Program B]** — …
• **[Program C]** — …
For the latest schedule of activities, see the Calendar.

Sources: Programs › [Program A] · Programs › [Program B] · Calendar

Visitor

Kailan ang susunod na activity sa Surigao del Norte?

LGMED AI Assistant

Ang susunod na nakatakdang activity sa Surigao del Norte ay ang **[Activity]** sa **[petsa]**, sa **[lugar]**.

Source: Calendar › [Activity]

Visitor

What is the rating of my mayor?

LGMED AI Assistant

I can't find that in the information LGMED has published. For questions about assessment results, please contact the Division directly.

Source: Contact › LGMED, DILG Caraga

The names in square brackets are placeholders. In the live assistant they are filled in from the actual published records.

Typical questions it can answer

Kind of question	Example	Answered from
Programs and projects	"What is [program]? Which LGUs are covered?"	Programs pages
Frontline services	"What are the requirements for [service]? How long does it take?"	Services page
News and advisories	"Are there new announcements this week?"	Announcements

Kind of question	Example	Answered from
Schedule	“Is there an activity in Butuan this month?”	Calendar
Accomplishments	“What did LGMED accomplish last week?”	Accomplishments
Figures	“How many LGUs are in Caraga?”	Statistics
Documents and reports	“Where can I download the [report/memo]?”	Documents, Reports
The office	“What are your office hours? Who do I contact?”	About, Contact

Languages

The assistant answers in the language the visitor uses: **English, Filipino, and Cebuano/Bisaya**. Hosted models handle all three well. Smaller local models are noticeably weaker in Filipino and Cebuano (see Section 6).

Later phases (for a separate decision)

- **Staff assistant (inside IMMS, signed-in users only):** summarise an incoming document, draft a routing note, draft the weekly accomplishment narrative, or ask “which LGUs have not yet submitted X?”. A person always reviews and approves what it drafts.
- Because this works with **internal, unpublished records**, it has stricter privacy requirements and is the main reason to consider local hardware (Section 7).

4. How it works

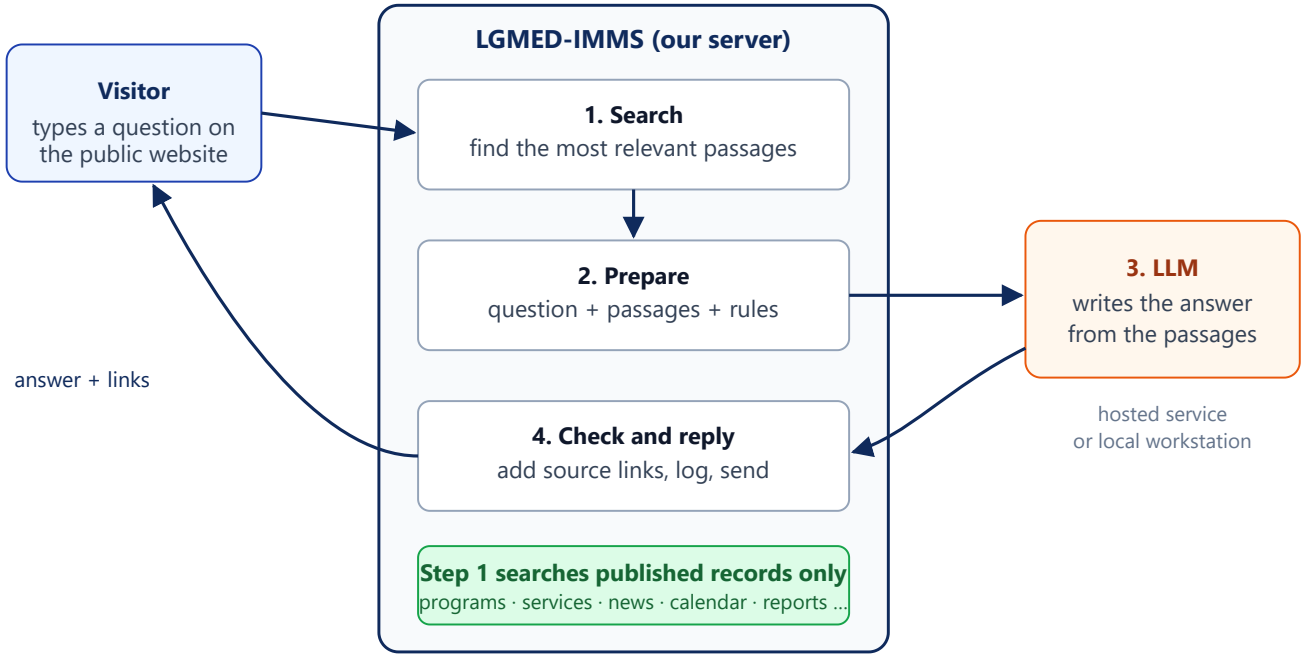


Figure 1. How one question is answered. Only published records go into step 1; internal modules are never searched.

Step by step

1. **Indexing (automatic, in the background).** Whenever a record is published, changed or unpublished in IMMS, it is cut into short passages and stored in a search index. Unpublishing a record removes it from the index straight away.
2. **Search.** When a question arrives, IMMS finds the handful of passages that best match its *meaning*, not just its words, so "requirements" also finds "documents needed".
3. **Prepare.** IMMS sends the LLM the question, those passages and fixed rules: answer only from the passages, never invent figures, names or dates, reply in the visitor's language, and keep it short.
4. **Answer.** The LLM writes the reply. IMMS adds links to the source pages, records the exchange in a log (Section 8), and shows the answer to the visitor.

Where it fits in IMMS

Part	What is added
New module `assistant`	The indexing, search and answering logic, plus a settings page for administrators.
Public website	A chat button and window on every public page.
Public Website settings	Switch the assistant on or off, edit its welcome message and disclaimer, and choose which kinds of records it may use.
Audit / Analytics	A log of questions, plus a report on what people ask most and which questions it could not answer, so the Division knows what information to publish next.

5. What it may and may not answer from

This is the most important safeguard. The public assistant reads **exactly what an anonymous visitor can already see on the website**, and nothing more.

✓ It uses	✗ It never uses
Published announcements, news and advisories	Incoming and Outgoing correspondence
Public program, project and activity pages	Unpublished drafts of any record
Frontline services (requirements, steps, fees, timelines)	Documents not marked public
Published weekly accomplishments	e-SIRA documents and signatures
Public statistics, reports and documents	User accounts, roles, personal data
Public calendar entries	Monitoring records not marked public
About, mandate, contact details, office hours	Audit logs, notifications, system settings

> **IMPORTANT:** The assistant does not get its own permission to see more. It reads through the same "is this published?" checks the public pages use. If a record would not appear on the website, the assistant cannot see it.

6. Three ways to run it

The “brain” (the LLM in step 3) can run in one of three places. Everything else (search, rules, logs, the chat window) stays inside IMMS in every option.

	A. Hosted AI service	B. Local AI workstation	C. Hybrid
What it is	IMMS sends the question and passages to a provider (e.g. Anthropic Claude, OpenAI, Google Gemini) over the internet	An open model (e.g. Llama, Qwen, Gemma) runs on a computer in our office	Hosted for the public assistant; local for a future staff assistant
Answer quality	Best, including Filipino and Cebuano	Good in English; weaker in Filipino and Cebuano; depends on the hardware	Best for the public; good for staff
Speed	2–6 seconds	3–20 seconds, depending on hardware and how many people ask at once	—
Up-front cost	None	Workstation purchase (Section 7)	Workstation, only when the staff phase is approved
Running cost	Pay per use; can be capped	Electricity; no per-use fee	Both, but small
Internet needed	Yes	No	Public part: yes
Where data goes	Published passages and visitor questions go to the provider	Nothing leaves the office	Only public data leaves
Maintenance	Very little	We update the models and drivers and look after the machine	Some
Time to pilot	About 3–4 weeks	About 5–7 weeks (includes procurement)	Phase 1 as A
Recommendation	**Start here** for the public assistant	When internal records are involved	**Long-term setup**

> **Why hosted is acceptable for the public assistant:** the passages it sends are already on our public website for anyone to read. The real privacy item is the **visitor's question**, which may contain a name or a personal concern. Section 8 covers how we handle that.

7. Hardware

If we use a hosted service (Option A)

No new hardware is needed. The computer that already runs IMMS handles the search and the chat. The heavy work happens at the provider. The only requirement is a stable internet connection on the server.

If we run the model ourselves (Option B or C)

An LLM runs mainly on the **graphics card (GPU)**, and the most important number is the GPU's **memory (VRAM)**: it limits how large, and therefore how capable, a model the machine can run. An ordinary office PC without a dedicated graphics card can technically run a small model, but too slowly for a public website.

Tier 1 — Pilot / demonstration TRY IT OUT

Form	A gaming or creator **laptop**, or a mid-range desktop
GPU	NVIDIA GeForce RTX 4060 / 5060 class, **8 GB VRAM** (laptop: RTX 4070 / 5070 Laptop, 8 GB)
Processor	Intel Core i7 / AMD Ryzen 7 (recent generation)
Memory (RAM)	32 GB
Storage	1 TB NVMe SSD
Runs	Small models (7–8 billion parameters), fine for testing in English, for 1–3 users at a time
Indicative price	₱75,000 – ₱110,000

Good for proving the idea and demonstrating it to management. A laptop is **not** suitable as the always-on public server: it runs hot under constant load and wears out faster.

Tier 2 — Office AI workstation RECOMMENDED FOR LOCAL

Form	**Desktop workstation** (tower), always on, in the server area
GPU	NVIDIA GeForce RTX 4090 (24 GB) or RTX 5090 (32 GB); acceptable lower option: RTX 4070 Ti Super / 5070 Ti (16 GB)
Processor	Intel Core i9 / Ultra 9, or AMD Ryzen 9
Memory (RAM)	64 GB
Storage	2 TB NVMe SSD (models are 5–40 GB each)
Power	1000 W 80+ Gold power supply, plus a **UPS** (1500 VA or higher)
Runs	Medium models (14–32 billion parameters) with good quality, for several users at once; handles the future staff assistant too
Indicative price	₱180,000 – ₱320,000 (the GPU is about half of the cost)

Tier 2 alternative — Apple Mac Studio / Mac mini QUIET, LOW POWER

Configuration	Apple M-series Max/Pro chip with **64 GB or more unified memory**, 1 TB SSD
Runs	Similar models to Tier 2, somewhat slower when many people ask at once, but silent and uses far less electricity
Indicative price	₱150,000 – ₱260,000

A good choice if there is no cooled server room. Check with ICT that macOS is acceptable for office equipment.

Tier 3 — Server-grade ONLY IF SCALED REGION-WIDE

Configuration	Rack server with a professional GPU (e.g. NVIDIA RTX PRO / L-series, 48 GB or more), 128 GB RAM, redundant power
When	Only if the assistant is offered to all DILG Caraga divisions or provincial offices, with heavy daily use
Indicative price	₱600,000 and up

> **About the prices:** these are indicative Philippine market ranges as of late 2026, for planning only. GPU prices change often. Final amounts must come from a canvass or PhilGEPS procurement following the usual rules, and ICT should confirm the specifications before purchase.

Electricity (local options)

A Tier 2 desktop draws about 100–150 W when idle and up to 600–700 W while answering. Left on all day with light use, plan for roughly **150–250 kWh a month**. A Mac Studio uses a fraction of that.

8. Data privacy and safeguards

Data Privacy Act (RA 10173)

- **Consult the Data Protection Officer (DPO)** before the pilot and before going public. A short privacy impact assessment is recommended.
- **Tell visitors.** The chat window shows a short notice: they are talking to an AI, the answer may be wrong, they should not type personal information, and questions are logged to improve the service. It links to the site's existing Privacy statement, which gets a new paragraph about the assistant.
- **Keep only what we need.** Logs keep the question, answer, sources used and time. IP addresses are kept only briefly, for abuse prevention. Logs are deleted after a set period (proposed: 90 days).
- **Hosted provider terms.** Use a business/API account whose terms say our data is **not used to train** the provider's models, with the shortest available retention.

Built-in safeguards

Safeguard	What it does
Published records only	The assistant cannot see internal modules (Section 5).
Answers must cite sources	Every answer links to its pages. If nothing relevant is found, it says so and points to the Contact page instead of guessing.
Fixed rules	It never gives legal rulings, never comments on politics or individual officials, never makes promises on behalf of the Division, and refuses unrelated tasks.
Disclaimer	“This is an AI assistant. Official information is on the linked pages and in issued documents.”
Personal data filter	Obvious personal details (mobile numbers, email addresses) in questions are masked before they are logged or sent out.
Abuse protection	A limit on questions per visitor per hour, a maximum question length, and reCAPTCHA (IMMS already supports it for sign-in) if abuse appears.
Spending cap	(Hosted) A monthly budget limit at the provider. When it is reached, the assistant turns itself off and shows the Contact page.
Kill switch	An administrator can turn the assistant off instantly from Public Website settings.
Human review	A staff member reviews a sample of conversations each week and flags wrong answers. These usually show missing or unclear published information, which is fixed at the source.

9. Costs

Option A — Hosted

- **Hardware:** none.
- **Usage:** providers charge by the amount of text processed. Each question costs a small fraction of a peso to a few pesos, depending on the model chosen and how long the answers are. The monthly total depends on how many questions visitors ask.
- **How we control it:** during the pilot we record the real cost per question, then set a monthly cap at the provider. A smaller, cheaper model from the same provider is usually enough for this kind of question-and-answer.
- **Payment:** providers bill by credit card or invoice in US dollars. Finance should confirm how this can be procured (for example, as a subscription service).

Option B — Local

Item	Amount
Tier 2 workstation and UPS	₱180,000 – ₱320,000 (one time)
Electricity	About 150–250 kWh a month
Software (open models, Ollama or vLLM)	Free (open-source licences; check each model's licence terms)
Staff time	A few hours a month for updates and monitoring

Development

Built in-house as a new IMMS module, using the existing codebase, the public website and its publishing rules. No outside developer or licence is required.

10. Implementation plan

Phase	Duration	Work	Done when
0. Approval	1–2 weeks	The team discusses this proposal; the Division Chief decides; the DPO is consulted; the hosted account or hardware purchase is arranged.	Signed approval, DPO advice on file
1. Build	2–3 weeks	Assistant module, indexing of published records, chat window, settings, logs, safeguards.	Works on a test copy of the site
2. Staff pilot	4 weeks	Only signed-in LGMED staff can use it. Staff ask real questions, mark answers right or wrong, and fill gaps in published content.	90% or more of the test questions answered correctly or correctly referred
3. Public beta	4–8 weeks	Switched on for the public with a "Beta" label. Weekly review of the logs and costs.	Stable costs, no serious incidents
4. Regular service	Ongoing	The Beta label is removed. A monthly report goes to the Chief: most-asked questions, unanswered questions, cost.	—
5. Staff assistant (optional)	Separate proposal	The internal assistant for drafting and summarising, on local hardware.	—

How we will measure success

- **Accuracy:** share of reviewed answers that are correct, or correctly say "I don't know". Target: 90% or more.
- **Usefulness:** the "Was this helpful?" thumbs up/down in the chat window.
- **Fewer routine enquiries** by phone, email and Facebook (compared with the months before).
- **Content gaps found:** topics people ask about that we had not published.
- **Cost per month** within the approved cap.

11. Risks and how we handle them

Risk	Likelihood	How we handle it
The assistant gives a wrong answer	Medium	It answers only from passages, must cite them, and says “I don't know” otherwise. The disclaimer, weekly review and staff pilot also help.
Someone tries to trick it into saying something improper	Medium	Fixed rules, refusal of off-topic requests, a length limit, logging and the kill switch. It has no access to anything that is not already public.
A visitor types personal information	Medium	A notice not to, masking of obvious personal details, short log retention, and provider terms that rule out training on our data.
Costs grow unexpectedly	Low	A monthly cap at the provider, a per-visitor limit, and automatic shut-off at the cap.
Provider outage or internet loss	Low	The chat window says the assistant is unavailable and points to the Contact page. The rest of the website keeps working.
Published information is outdated, so the answers are too	Medium	The assistant is only as good as what we publish. The log of unanswered questions shows what needs updating.
The public mistakes an AI answer for an official statement	Low	A visible AI label, the disclaimer, and links to official pages in every answer.

12. Decisions needed from the team

1. **Go ahead with Phase 1 (public assistant, published information only)?** ☐ Yes ☐ No ☐ Revise
2. **Where should the AI run?** ☐ A. Hosted (recommended) ☐ B. Local workstation ☐ C. Hybrid
3. **Hardware (if B or C):** ☐ Tier 1 pilot laptop ☐ Tier 2 workstation ☐ Mac alternative
4. **Monthly spending cap** (if hosted): ₱ _____
5. **Who reviews conversations weekly?** _____
6. **Who is the focal person for the DPO consultation?** _____
7. **Explore the staff assistant later (Phase 5)?** ☐ Yes ☐ Not now

Approval

Role	Name	Signature	Date
Prepared by			
Reviewed by (ICT)			
Reviewed by (Data Protection Officer)			
Approved by (Division Chief, LGMED)			

13. Glossary

AI assistant / chatbot	A chat window where people type questions and get written answers.
LLM (large language model)	The AI technology that understands and writes language, e.g. Claude, GPT, Gemini, Llama.
Hallucination	When an LLM states something false with confidence. Answering only from our passages is how we prevent it.
RAG (retrieval-augmented generation)	Searching our own records first, then having the LLM answer only from what was found.
Index	A searchable copy of the published records, cut into short passages, kept up to date automatically.
Hosted service / API	An LLM run by a provider on their servers, used over the internet and paid per use.
Local / open model	An LLM whose files can be downloaded and run on our own computer, e.g. Llama, Qwen, Gemma.
Parameters (e.g. “8B”, “32B”)	A rough measure of a model's size. Larger models are usually more capable but need more GPU memory.
GPU / VRAM	The graphics card and its memory. The main hardware an LLM runs on; VRAM limits the model size.
Ollama / vLLM	Free programs for running open models on our own hardware.
DPO	Data Protection Officer, who advises on compliance with the Data Privacy Act (RA 10173).